LI Jiheng

Email: ljh20050406@163.com
Hongshan District, Wuhan, Hubei, China

EDUCATION

Huazhong University of Science and Technology 09/2024 – 06/2028
Bachelor of Engineering in Intelligence Science and Technology GPA: 4.28/5.00

- **Modules:** Machine Learning (97), Discrete Mathematics (95), Machine Learning Course Design (94), Numerical Analysis (93), Computer Systems (92), Algorithm Design and Analysis (90)

RESEARCH INTERESTS

- Interested in reliable and adaptive AI systems that can perceive, reason, and act under noisy, incomplete, or shifting visual information, with particular attention to how perception quality affects downstream decision-making.
- Interested in learning transferable representations and dynamics from large-scale visual interaction data, especially how data quality, temporal context, and cross-domain variation influence generalization beyond a single environment or task.
- Interested in robust perception over time rather than isolated-frame accuracy, including whether historical observations can sustain reliable state estimates when an agent or robot moves from a well-observed region into a visually unreliable one.
- Interested in multimodal and computer-use agents, particularly how visual information can be represented, verbalized, and selectively used to support action, and how evaluation design can distinguish genuine capability gains from prompt or system-level confounds.

RESEARCH & PROFESSIONAL EXPERIENCE

Are Multimodal Skills Necessary for GUI Agents?—Analysis and Text-Only Skill Distillation
07/2026 – Present

Independent Researcher

- Investigate whether multimodal skills are genuinely necessary for GUI agents by controlling prompt-level confounds in OSWorld. Refactored four prompt scaffolds into independently switchable modules and ran a 360-task ablation, separating gains from scaffolding and skill knowledge; prompt scaffolding accounts for about 58% of the reported improvement.
- Trace performance differences to concrete failure mechanisms, including output parsing, premature self-termination, and execution-harness effects. The analysis shows that many failures persist across conditions, motivating a shift from capability-only comparisons toward more controlled and mechanism-based evaluation.
- Audit the text-only condition at the implementation level and identify information loss and visual-cue misalignment between reference images and the runtime UI, showing that the existing comparison cannot cleanly isolate the value of multimodal skills.
- Currently extend this work toward text-mediated computer-use agents by converting task-relevant visual information into structured textual descriptions that can support agent actions. Develop Skill

Verbalization and an offline discriminability check using hard negatives, with the longer-term goal of identifying when text is sufficient and when multimodal information should be invoked.

Task-Specific Visual Perception for Mobile Robots 07/2026 – 08/2026

Research Intern

- Adapted and trained the open-source YOLOv8n detector for task-specific perception of humans, Unitree Go2 quadrupeds, and Scout robots on indoor and outdoor ZED RGB data, evaluating cross-scene robustness rather than treating YOLOv8n as a newly developed model.
- Built an end-to-end data and deployment pipeline covering video collection, annotation, training, evaluation, and Jetson inference, and integrated live ROS2 streams through rosbridge in Docker to test the perception stack under robot-side operating conditions.
- Reduced mean inference latency by 22.6% versus YOLO11n (25.22 to 19.53 ms), increased throughput from 30.42 to 36.32 FPS, and achieved about 90% pooled frontal human detection at 6 m. Based on observed failure cases, further explored a temporal-recovery question: whether detection history from a reliable region can help sustain perception after a target moves into an unreliable region, using prior detections together with LiDAR-based recovery cues as a follow-up direction.

DynamiX: General Inverse Dynamics for World Modeling 03/2026 – 06/2026

Research Intern

- Contributed to a cross-game inverse-dynamics project that seeks to infer action/control signals from gameplay videos and learn transferable action-visual dynamics, with primary responsibility for data-engine construction, filtering, and quality control across heterogeneous video sources.
- Built the GameMotion-A data engine to support cross-game inverse dynamics modelling, processed approximately 5,800 hours of YouTube gameplay footage, and introduced automated frame-level quality checks to convert unstructured gameplay videos into trainable data resources.
- Develop a multi-stage feedback and filtering pipeline based on Qwen3-VL-30B-A3B, establishing clear rules around perspective, host, comments, transitions, UI/HUD, and generating higher-quality segments for action modeling to improve data purity, consistency, and action relevance.

TECHNICAL SKILLS

- Multimodal Data Engine; General Inverse Dynamics; VLM-based Filtering (Qwen3-VL); Action Modeling; YOLOv8n; ROS2; Jetson Inference; Docker; LiDAR-based Recovery; GUI Agents; OSWorld; Ablation Study; Reproducible Experimentation; Python; PyTorch.